

Aziz Mohammad

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EDUCATION

University of Illinois Chicago, Chicago, IL

Aug 2025 – May 2027 (Expected)

BS in Mechanical Engineering

GPA – 3.8/4.0 | Technical GPA – 3.86/4.0 | Dean's List

Relevant Coursework: Compressible Flow, Heat Transfer, Fluid Mechanics, Thermodynamics I & II

FE Mechanical Exam — Planned Winter 2026–2027

ENGINEERING EXPERIENCE

Wilo USA, Systems Engineering Intern, Rosemont, IL

May 2026 – August 2026

- Contributed to the development and standardization of a modular hydronic pump package product family spanning 30–230 tons and 20+ configurations for commercial air-cooled chiller systems
- Developed and refined SolidWorks assemblies and 2D package layouts integrating pumps, VFDs, tanks, piping, electrical equipment, and structural skid components across multiple system configurations.
- Built and maintained configuration matrices linking system capacity to pump/motor selections, VFDs, tank sizing, water-treatment components, and package requirements, enabling repeatable designs across the product family.
- Evaluated supplier drawings, equipment weights, and component specifications while cross-referencing metric and U.S. hardware to support package design, sourcing, and manufacturability.

University of Illinois Chicago, Research Assistant at Intelligent Multiphase Systems Lab

March 2026 – Present

- Developing a precision flow-boiling test platform to study multiphase heat transfer and interfacial transport for high-performance electronics cooling.
- Designed and prototyped SolidWorks experimental hardware incorporating fluid routing, mechanical interfaces, instrumentation, and manufacturability using FDM/SLA additive manufacturing.
- Integrated flow measurement, DAQ hardware, sensors, signal wiring, and power connections; supported calibration, troubleshooting, system validation, and repeatable test-data acquisition.

PROJECTS AND LEADERSHIP

Society of Automotive Engineers (SAE), Member

Aug 2025 – Present

- Designed and analyzed a custom Formula SAE push bar using SolidWorks and ANSYS Workbench, evaluating structural performance and weight through iterative FEA.
- Designed and fabricated a welding jig for the Formula SAE tubular space frame to support suspension hard-point alignment and chassis symmetry.

American Society of Mechanical Engineers (ASME), Member

Aug 2025 – Present

- Applied mechanical design and CAD modeling in team projects; participated in technical presentations and networking events focused on engineering and manufacturing practices.

The Scythe — Structural Design and FEA | SolidWorks, ANSYS Workbench

- Designed a cantilever traffic-signal pole and evaluated safety factor and deformation under gravity, equipment loads, and wind pressure using ANSYS Static Structural.

3D-Printed Gear Assembly | SolidWorks, FDM

- Designed and printed a planetary-gear fidget toy, applying parametric assembly constraints, backlash adjustments, and clearance fits for FDM fabrication.

SKILLS

- **Software & Analysis:** SolidWorks, Creo, AutoCAD, ANSYS Workbench, MATLAB, Python, Excel, Google Workspace, Microsoft Office; FEA, engineering data analysis, workflow automation
- **Engineering:** SLA/FDM 3D Printing, CNC, Manual Machining, GD&T, DFM, Drawings, BOM Creation DAQ, flow measurement, sensors and signal wiring